The Kalman Filter

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Introduction

The Kalman filter is the workhorse algorithm for state-space models with linear-Gaussian observation and transition equations. It recursively combines the previous state estimate with each new observation to produce an updated state estimate together with its uncertainty, in a closed-form Gaussian update that is provably optimal under the linear-Gaussian assumptions. Originally developed for aerospace tracking, it now underpins virtually every classical time-series filtering application — economic indicators, signal processing, target tracking, financial volatility estimation.

Prerequisites

A working understanding of state-space models, linear algebra, and the multivariate Normal distribution.

Theory

For a linear-Gaussian state-space model with observation $y_t = Fx_t + v_t$ and transition $x_t = Gx_{t-1} + w_t$, the Kalman filter alternates two steps at each time:

Predict:

$$x_{t|t-1} = Gx_{t-1|t-1}, \qquad P_{t|t-1} = GP_{t-1|t-1}G^\top + W.$$

Update:

$$x_{t|t} = x_{t|t-1} + K_t(y_t - Fx_{t|t-1}), \qquad P_{t|t} = (I - K_tF)P_{t|t-1},$$

with Kalman gain $K_t = P_{t|t-1}F^\top(FP_{t|t-1}F^\top + V)^{-1}$.

The smoother (Rauch-Tung-Striebel) is a backward recursion that combines forward filter estimates with future observations to produce smoothed estimates $x_{t|T}$. Non-linear extensions include the extended Kalman filter (linearisation around the current estimate) and the unscented Kalman filter (sigma-point sampling); particle filters handle fully non-Gaussian and non-linear models via Monte Carlo.

Assumptions

Linear Gaussian observation and transition equations with known parameters (F, G, V, W). Parameters can be estimated by MLE on the innovations.

R Implementation

```r
library(dlm)

# Local-level model on Nile
mod <- dlmModPoly(order = 1, dV = 15100, dW = 1470)
filt <- dlmFilter(Nile, mod)

plot(Nile)
lines(dropFirst(filt$m), col = "red", lwd = 2)
```

```r
# Smoother
smo <- dlmSmooth(filt)
lines(dropFirst(smo$s), col = "blue", lwd = 2)
```

Output & Results

`dlmFilter()` returns the filtered state estimates ($x_{t|t}$) and their covariances. `dlmSmooth()` produces the retrospective smoothed estimates ($x_{t|T}$). Plotting both on the original series shows the filter's adaptation in real time and the smoother's retrospective best estimate.

Interpretation

A reporting sentence: "The Kalman filter applied to the Nile river-flow series produced one-step-ahead forecasts (filtered states) that adapt to each new observation; the smoothed estimate, using the full series in retrospect, identified a sharp level drop near 1899 that the forward filter takes several years to track." Distinguish filtered (real-time) from smoothed (retrospective) estimates in any report.

Practical Tips

- For non-linear models, use the extended Kalman filter (`KFAS::EKF` or hand-coded) for mild non-linearity, the unscented Kalman filter for stronger non-linearity, or particle filters (`SMCSamplers`, `SMC2`) for arbitrary non-linear / non-Gaussian models.
- Uncertainty propagates naturally through both predict and update steps; report state estimates with their estimated standard errors.
- The log-likelihood of the model can be computed from the innovations (`dlmLL` in `dlm`) and used for MLE-based parameter estimation; the Kalman filter is the engine, MLE is the parameter-fitting wrapper.
- The choice of initial state $x_{0|0}$ and initial variance $P_{0|0}$ matters for the first few time steps; diffuse initialisation ($P_{0|0}$ very large) is the standard choice when no prior information is available.
- For extremely high-dimensional state spaces (large meteorological models), the ensemble Kalman filter approximates the covariance with a low-rank ensemble; standard Kalman fails in that setting.
- Pair filtering with diagnostic checks on the innovations (one-step-ahead residuals); they should be approximately white noise under correct model specification.

R Packages Used

`dlm` for `dlmFilter()`, `dlmSmooth()`, and `dlmLL()`; `KFAS` for fast filtering with non-Gaussian observations; `FKF` for the fast Kalman filter implementation; `bsts` for Bayesian structural time-series alternatives; `nimble` for hand-built non-linear filters via probabilistic programming.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics